



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 12

Number

1 2 1 0 7 X

UNIT TEST - 1, JULY - 2024

Time Allowed : 1.30 Hours

PHYSICS

[Max. Marks : 35]

PART - I

Choose the correct Answer.

5x1=5

- Which charge configuration produces a uniform Electric Field?
(a) Point charge (b) Uniformly charged infinite line
(c) Uniformly charged infinite plane (d) Uniformly charged spherical shell
- A Capacitor of $50\mu\text{F}$ is charged to 10 volts Its energy in Joules is
(a) 2.5×10^{-3} (b) 5×10^{-3} (c) 10×10^{-4} (d) 2.5×10^{-4}
- A Internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10Ω is
(a) 0.2Ω (b) 0.5Ω (c) 0.8Ω (d) 1.0Ω
- n equal resistors are first connected in series and then in parallel. The ratio of the equivalent resistance in two case is
(a) n (b) $\frac{1}{n^2}$ (c) n^2 (d) $\frac{1}{n}$
- The vertical component of Earth's magnetic field at a place is equal to the horizontal component
What is the value of Angle of dip at this place?
(a) 30° (b) 45° (c) 60° (d) 90°

PART - II

Answer any Three questions and Question number 10 is compulsory.

3x2=6

- What is an Equipotential Surface?
- State : Gauss Law.
- Define Temperature Coefficient of resistance.
- State Fleming's Left Hand rule.
- Find the heat energy produced in a resistance of 10Ω when 5 A current flow through it for 5 minutes.



PART- III

Answer any three questions and question No. 15 is compulsory.

3x3=9

11. Derive an expression for Electric Potential due to a point charge
12. Discuss the conversion of Galvanometer into an Ammeter.
13. Explain Cells in Series.
14. State Kirchhoff's Junction rule and Loop rule.
15. A parallel plate Capacitor has square plate of side 5 Cm and separated by a distance of 1 mm. Calculate the Capacitance of this capacitor.

PART - IV

Answer all the questions.

3x5=15

16. a) Explain in detail the Construction and Working of a Van de Graff Generator.

(OR)

- b) Explain in detail the effect of Dielectric placed in a parallel plate capacitor, When the capacitor is disconnected from the battery.

17. a) Obtain the condition for bridge balance in wheatstone's bridge.

(OR)

- b) Describe the Microscopic model of current and Obtain Microscopic form of Ohm's law.

18. a) Discuss the Working of Cyclotron in detail.

(OR)

- b) Derive the expression for the force on a current - carrying Conductor in a Magnetic Field.

b) Derive an expression for electric dipole along axial line, electric field due to